

Johan Naresh

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EXPERIENCE

Quantitative Researcher

Feb 2026 - Present

Wilmot ML

- Developed a real-time nowcasting framework for industrial production integrating **10+** macro data sources across varying time frequencies using a **Kalman filter** state-space model, with vintage-controlled backtesting ensuring real-time data integrity
- Formulated a state-space modeling pipeline separating observation and transition dynamics, and integrated **ridge regression** to regularize high-dimensional macroeconomic feature sets and improve coefficient stability under multicollinearity
- Designed a rigorous validation framework using permutation testing, block cross-validation, and expanding-window evaluation to isolate true signal contribution, achieving a **5.4%** out-of-sample RMSE reduction over baseline models

Data Science Research Analyst

Dec 2025 - Present

(Startup) Sephira Institute - Public Sentiment Team

- Built end-to-end **Python ETL pipelines** for global equity and sentiment data, engineering a country-month panel with aligned time indices, imputation, and feature scaling for cross-sectional and **time-series modeling**
- Identified a consistent peak lag between in-house sentiment signals and equity returns across markets and sectors using rolling correlations, log-return transformations, and OLS residualization; relationship held across multiple economic regimes
- Designed and backtested a monthly sentiment-driven strategy with strict no look-ahead bias, achieving **11.9% CAGR** vs 9.0%, Sharpe 0.76 vs 0.49, beta 0.73; validated robustness via volatility, drawdown, and CAPM alpha diagnostics

Quantitative Developer

Dec 2025 - Present

Wat Street - Quant Design Team

- Created a high-throughput automated **Python** pipeline to process large-scale financial news streams, leveraging **MinHash and locality-sensitive hashing (LSH)** to eliminate near-duplicate articles at scale before embedding-based clustering.
- Developed a **time-aware semantic clustering framework** that dynamically tightens similarity thresholds for temporally proximate articles, isolating true market-moving events from redundant news flow.

Software Engineer

Apr 2025 - Mar 2026

Marble Investments

- Engineered asynchronous, semaphore-controlled API ingestion pipelines in Python (asyncio + httpx) with rate-limit handling, retry logic, and structured validation, reducing acquisition time by **80%** and enabling vectorized Pandas transformations
- Built indexed **MongoDB** architectures with upsert refresh logic to enable scalable ETL processing and low-latency queries.
- Developed a **Next.js + TypeScript** internal analytics dashboard streaming live financial KPIs, enabling real-time portfolio monitoring, exploratory data analysis, and investment decision support.

PROJECTS

🔗 Real-Time Gym Rep Counter | C++, Node.js, WebSockets, Supabase

- Built low-latency rep counting in **C++** using accelerometer and gyroscope streams and defining rep-detection logic (filtering, peak/valley detection, debouncing, timing constraints), with real-time sensor ingestion .
- Developed a **full-stack pipeline** where, **Node.js** backend ingests high-frequency JSON sensor streams, validates and buffers events, and sends updates via **WebSockets** to a responsive HTML/CSS/JS dashboard with real-time state synchronization.
- Connected the dashboard to **Supabase** for live session state (counts, metrics) and added **computer-vision** features.

📱 ESG Mobile App - Finalist SMAC Competition | Dart, C++, CMake, Gemini API

- Built a **Flutter** Android ESG-scoring app with barcode scanning and image capture inputs, piping multi-modal product data to the Gemini API to generate per-criterion ESG scores and natural language explanations rendered dynamically in-app.
- Implemented a searchable history screen with card-based UI, persisting user preferences and scan history via local device storage

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C++, C, HTML/CSS, Java, Dart, Bash

Frameworks & Libraries: React, Next.js, Flask, Pandas, TensorFlow, BeautifulSoup, Selenium, yfinance

Tools & Platforms: Git, GitHub Actions, Linux, PostgreSQL, MongoDB, Firebase, Vercel, Power BI, Excel, APIs (REST, LLMs)

EDUCATION

University of Waterloo

Computer Science and Finance, Bachelor of Computing and Financial Management (Honours Co-op)

- GPA:** 3.94/4.00 | *Awarded President's Scholarship of Distinction*
- Relevant Coursework:** Algorithm Design, Data Abstraction, Financial Markets & Data Analytics, Linear Algebra